

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

Duration :60 Minutes

Total Marks: 40

Code of Conduct

I sanjay kumar soy(192311151) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

sanjay kumar soy

Signature of the student:

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- | | | | |
|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |

5. **Agriculture** **Smart** **Farming** **Problem**
 A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

ANSWER

1. Retail Data Optimization Problem – Big Data Architecture

A suitable solution is a real-time big data architecture that collects data from physical stores and the online platform, processes it in real time and batch mode, stores it in a scalable data lake, and applies analytics and machine learning for forecasting and personalization.

1. Data Sources

The architecture collects data from:

- Physical stores: POS transactions, inventory scanners, billing systems
- Online platform: website/app clicks, searches, purchases, cart activity
- Customer data: profiles, preferences, purchase history
- Inventory systems: stock levels, product movement, warehouse data
- External data: holidays, weather, promotions and market trends

2. Data Ingestion

Tools such as Apache Kafka can collect and stream large volumes of data in real time.

For example:

Customer purchases a product → POS generates an event → Kafka receives the event → processing system analyzes it.

For batch data, tools such as Apache Sqoop or cloud-based ingestion services can be used.

3. Data Storage

A scalable **Data Lake** can store both structured and unstructured data.

- HDFS – distributed storage
- Amazon S3 / Azure Data Lake / Google Cloud Storage – cloud-based alternatives
- Apache Hive – querying and analyzing large datasets

This allows the company to store sales records, customer activity, inventory information, and other data in one central location.

4. Real-Time Data Processing

Apache Spark Streaming or **Apache Flink** can process incoming data in real time.

This can be used for:

- Real-time inventory updates

- Detecting low-stock products
- Monitoring sales
- Tracking customer activity
- Generating immediate alerts

For example, when a product is sold in a physical store, the inventory system can immediately update the available stock.

5. Demand Forecasting

Historical sales data can be analyzed using machine learning models such as:

- Linear Regression
- Random Forest
- XGBoost
- ARIMA / time-series models
- LSTM for more advanced forecasting

The model can predict future demand based on factors such as previous sales, seasonality, promotions, holidays and customer behavior.

Example:

Historical data shows that umbrella sales increase during the rainy season → the model predicts higher future demand → the company increases inventory before the season.

6. Personalized Marketing

Customer behavior can be analyzed to generate personalized recommendations.

Models such as:

- Collaborative Filtering
- Content-Based Filtering
- Recommendation Systems
- Clustering (K-Means)

can be used.

For example:

A customer frequently purchases sports shoes → the system identifies similar products → the customer receives personalized recommendations or offers.

7. Analytics and Visualization

Tools such as Power BI, Tableau or Apache Superset can be used to create dashboards.

Managers can monitor:

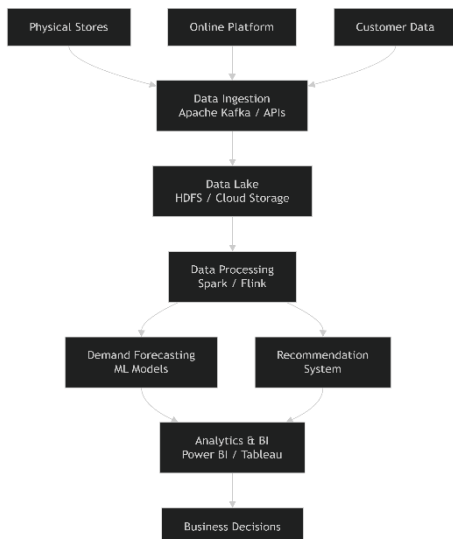
- Current inventory
- Sales trends
- Product demand
- Customer behavior
- Revenue
- Forecasted demand

8. Security

The architecture should include:

- Authentication and authorization
- Data encryption
- Role-based access control
- Secure APIs
- Customer privacy protection
- Regular backups and auditing

Overall Architecture



2. Social Media Fraud Detection Problem

A social media platform generates **billions of posts, comments, likes, shares, and user activities every day**. Since fake accounts and misinformation can spread very quickly, the platform needs a **scalable and real-time big data system**.

1. Data Sources

The system collects data from:

- User posts and comments
- Likes, shares and follows
- Login and device information
- IP address and location patterns
- Account creation and activity history
- Links, images and videos

2. Data Ingestion

Apache Kafka can be used to collect billions of events continuously.

For example:

User posts → Kafka receives the event → event is sent to the processing system.

Kafka is suitable because it is **distributed, fault-tolerant and capable of handling high-speed data streams**.

3. Stream Processing

Frameworks such as **Apache Spark Streaming** or **Apache Flink** process the incoming data in real time or near real time.

They can detect patterns such as:

- Thousands of likes from newly created accounts
- Abnormally high posting frequency
- Multiple accounts using the same IP/device
- Sudden sharing of suspicious links
- Coordinated activity between accounts

If suspicious behavior is detected, the system can immediately generate an alert or temporarily restrict the account.

4. Machine Learning for Fake Account Detection

Machine learning models can classify accounts as **genuine or suspicious**.

Useful features include:

- Account age
- Number of followers/following
- Posting frequency
- Number of likes and shares
- Login patterns
- IP/device similarity
- Network connections

Models such as **Random Forest, Decision Tree, Logistic Regression or XGBoost** can be trained using previously identified fake and genuine accounts.

5. Machine Learning for Misinformation Detection

For misinformation detection, the system can analyze:

- Text content
- Keywords
- URLs
- User/source credibility
- Sharing patterns
- Historical information

NLP (Natural Language Processing) models can analyze the text and classify content as potentially misleading or trustworthy.

6. Scalable Storage

Processed data can be stored in a distributed system such as:

- **HDFS**
- **Amazon S3**
- **Data Lake**

Historical data can then be used to retrain and improve the machine learning models.

7. Overall Architecture



8. Advantages

- **Real-time detection** of suspicious activity
- Handles **billions of events** using distributed processing
- Machine learning improves detection accuracy
- Scalable by adding more processing nodes
- Historical data can be used to continuously improve the models
- Helps reduce the spread of fake accounts and misinformation

3. Government Data Platform Problem – Governance and Data Management Strategy

A national government data platform must integrate **census, economic, and geospatial data** from multiple departments. Since different agencies need to share this information, the system should provide **standardization, interoperability, security, privacy, and proper data governance**.

1. Data Governance

Create a central **Data Governance Committee** to define policies and standards for all departments.

Its responsibilities include:

- Defining data ownership and responsibilities
- Establishing data-sharing policies
- Defining data quality standards
- Managing privacy and security requirements

- Monitoring compliance

Each major dataset should have a **Data Owner** responsible for its accuracy, access and maintenance.

2. Data Integration

Data from different departments may use different formats and structures. Therefore, common standards should be established.

For example:

Census Department —┐

Economic Department —┐

Geospatial Department └┐

Other Agencies ———┐

↓

Data Integration Layer

↓

National Data Platform

Use technologies such as:

- **APIs** for controlled data sharing
- **ETL/ELT tools** for data integration
- **Apache Kafka** for real-time data exchange when required
- Common formats such as **JSON, CSV and XML**
- Standard metadata and data schemas

3. Data Storage and Management

A centralized **data lake or data warehouse** can store integrated datasets.

- Data Lake → stores raw structured and unstructured data
- Data Warehouse → stores cleaned and structured data for reporting
- Metadata catalog → records information about datasets, sources, owners and formats

This makes it easier for departments to discover and reuse available data.

4. Interoperability

To ensure different government departments can communicate and exchange data:

- Use common data formats and schemas.
- Define standard APIs.
- Use common identifiers for entities such as people, organizations and locations.
- Maintain a common metadata standard.
- Follow standardized geospatial formats for geographic data.

This prevents each department from maintaining isolated data systems.

5. Data Quality Management

Data should be regularly checked for:

- Accuracy
- Completeness
- Consistency
- Duplicate records
- Timeliness

For example, if two departments have different values for the same economic indicator, data-quality rules should identify the conflict and determine the authoritative source.

6. Privacy and Security

Since government datasets may contain sensitive citizen information, strong security controls are required.

- Encryption during storage and transmission
- Role-Based Access Control (RBAC)
- Multi-factor authentication
- Data masking and anonymization
- Secure APIs
- Audit logs
- Regular security assessments

Only authorized users should access sensitive information.

7. Data Privacy

Personal information should be collected and used only for legitimate purposes.

Important measures include:

- Data minimization
- Anonymization/pseudonymization
- Purpose-based access
- Retention and deletion policies
- Consent or appropriate legal authority where required

8. Overall Strategy

Government Departments



Data Collection



ETL / APIs / Kafka



Data Integration & Standardization



National Data Lake / Warehouse



Data Catalog + Metadata



Analytics & Data Sharing



Government Departments

Security and governance are applied across all layers.

4. Banking Fraud Detection Problem – Big Data Pipeline

A bank processes millions of transactions every minute, so the system must detect suspicious transactions **quickly and accurately**. A suitable solution is a **real-time big data pipeline** that combines streaming data with historical transaction data and machine learning.

Data Sources

The system collects:

- ATM transactions
- Credit/debit card transactions
- Online and mobile banking
- UPI and payment transactions
- Customer account information
- Device, IP and location information
- Historical fraud records

Data Ingestion

Apache Kafka can be used to collect transaction events in real time.

For example:

Customer makes a ₹50,000 transaction → transaction is sent to Kafka → Kafka streams it to the fraud detection system.

Kafka is suitable because it can handle high-volume, high-speed data with low latency.

Stream Processing

Apache Flink or **Spark Streaming** can process transactions immediately.

The system can check:

- Transaction amount
- Transaction location
- Transaction frequency
- Device information
- Previous transaction behavior
- Multiple transactions within a short time

For example, if a customer's card is used in Chennai and then suddenly in London within five minutes, the transaction can be flagged as suspicious.

Historical Data Storage

Historical transactions can be stored in:

- HDFS

- Amazon S3 / Data Lake
- **Data Warehouse**

This historical data is useful for identifying customer behavior and training machine learning models.

Machine Learning Algorithms

The system can use both supervised and unsupervised learning.

Supervised algorithms:

- Random Forest
- XGBoost
- Logistic Regression

These models are trained using previously labelled transactions as **fraudulent or legitimate**.

Unsupervised algorithms:

- Isolation Forest
- Clustering

These can identify unusual transaction patterns even when labelled fraud data is limited.

Real-Time Fraud Detection

The ML model generates a **fraud/risk score** for every transaction.

Transaction



Kafka



Flink / Spark Streaming



Feature Extraction



ML Fraud Detection Model



Risk Score



Low Risk —→ Approve

High Risk —→ Block / Alert / Manual Review

For example:

Normal transaction → Low risk → Approve

Unusual location + very high amount + abnormal frequency → High risk → Block/verify

Low Latency and Accuracy

To achieve low latency:

- Use **Kafka** for real-time ingestion.
- Use **Flink/Spark Streaming** for stream processing.
- Keep frequently required customer information in a fast database/cache.
- Use optimized ML models for quick prediction.

To improve accuracy:

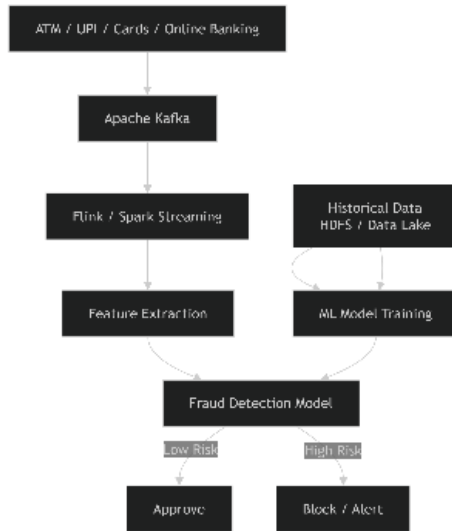
- Continuously train models using new fraud cases.
- Combine transaction rules with ML predictions.
- Regularly monitor false positives and false negatives.

Security

The system should also use:

- Encryption
- Authentication and authorization
- Role-based access control
- Audit logs
- Secure communication
- Customer data protection

Overall Architecture



5.Agriculture Smart Farming Problem –

A smart farming system collects large amounts of data from **soil sensors, weather stations, drones, satellites and crop monitoring systems**. Since this data comes from distributed sources and may be generated continuously, a big data architecture can help farmers make better decisions about irrigation, fertilizers, crop health and harvesting.

Data Sources

The system collects data from:

- **Soil sensors** – moisture, temperature, pH and nutrient levels
- **Weather sensors** – temperature, rainfall, humidity and wind speed
- **Drones** – crop images and plant health information
- **Satellites** – field and vegetation information
- **Farm equipment** – irrigation and machinery data
- **Historical crop data** – previous yields and farming records

Data Collection and Ingestion

IoT sensors continuously generate data. Technologies such as MQTT can collect sensor data, while Apache Kafka can be used to handle large-scale real-time data streams.

Data Storage

The collected data can be stored in a **distributed data lake**.

Suitable technologies include:

- HDFS
- Amazon S3 / Cloud Storage
- **Apache Hive** for querying large datasets

Both sensor data and unstructured data such as drone images can be stored for further analysis.

Real-Time Data Processing

Apache Spark Streaming or Apache Flink can process sensor data in real time.

For example:

Soil moisture becomes too low → sensor sends data → streaming system processes it → system detects dry soil → irrigation alert is generated.

This allows farmers to respond quickly to changing field conditions.

Data Analytics

Different analytics techniques can be used for precision agriculture.

Descriptive analytics:

- Shows current soil moisture and weather conditions.
- Provides reports about previous crop performance.

Predictive analytics:

- Predicts crop yield.
- Predicts irrigation requirements.
- Predicts possible pest or disease outbreaks.
- Forecasts weather-related risks.

Prescriptive analytics:

- Recommends how much water or fertilizer should be used.
- Suggests the best time for irrigation or harvesting.

Machine Learning

Machine learning models can analyze historical and real-time data.

Examples include:

- Random Forest – crop yield and disease prediction

- Decision Tree – farming decision classification
- Regression models – yield prediction
- K-Means clustering – grouping fields based on soil characteristics
- CNN – analyzing drone images for crop disease detection

7. Decision-Making

Analytics can provide farmers with useful recommendations.

For example:

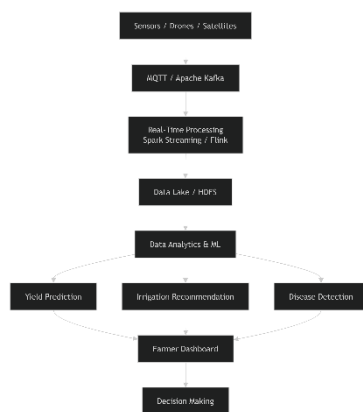
Low soil moisture + high temperature → Increase irrigation

Unhealthy crop pattern in drone image → Check for disease

Weather forecast + crop maturity → Recommend suitable harvesting time

Soil nutrient deficiency → Recommend appropriate fertilizer

8. Overall Architecture



9. Benefits

- Improves crop yield through data-driven decisions
- Reduces water usage through smart irrigation
- Reduces fertilizer usage by identifying actual soil requirements
- Detects crop diseases early
- Reduces farming costs
- Provides real-time field monitoring
- Supports better harvesting decisions